

Basic Notations of Probability Theory

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Sigma field and Measurable space

A **sample space** Ω is simply any nonempty abstract set. The physical meaning is the collection of all relevant possible outcomes of an experiment or a physical process.

A **σ -field/algebra** $\mathcal{F}$ in Ω is a nonempty collection of subsets of Ω that contains Ω and is closed under complementation and countable unions.¹

Endowed with a σ -field $\mathcal{F}$ in it, the space Ω is known as a **measurable space**, usually denoted by $(\Omega, \mathcal{F})$. A subset $A \subset \Omega$ is called **measurable** if $A \in \mathcal{F}$.

Examples of a σ -field in Ω include $\{\emptyset, \Omega\}$, and $2^\Omega =$ the collection of all subsets of Ω (the power set of Ω).

¹Note by this definition, a σ -field has to be closed under other ordinary set operations such as union, intersection, difference, and symmetric difference, as long as the operands (sets) involved are at most **countably** infinite (could of course be finite). Note however, it is not required to be closed under uncountable set operations. The σ -field is a mathematical model for the collection of all the events for which we would like, and are able, to assign probabilities in a manner consistent with our intuition/expectation.

Generated σ -fields

For any collection $\mathcal{C}$ of subsets in Ω , there is a smallest σ -field containing $\mathcal{C}$. This σ -field is called the σ -field generated by $\mathcal{C}$, denoted by $\sigma(\mathcal{C})$. Many interesting σ -fields are thus defined. For instance, **the Borel σ -field over the real line**, $\mathcal{B} \equiv \mathcal{B}(\mathbb{R})$, is defined as the σ -field generated by all open intervals in $\mathbb{R}$. Generally **the Borel σ -field** in a topological space is the σ -field generated by all the open sets (or by all closed sets).

Another useful concept is the σ -field generated by a mapping. If f is an arbitrary mapping from Ω to the measurable space $(T, \mathcal{T})$, then it is easy to show $f^{-1}\mathcal{T} = \{f^{-1}B; B \in \mathcal{T}\}$, where $f^{-1}B \triangleq \{\omega \in \Omega; f(\omega) \in B\}$, is a σ -field in Ω . This σ -field is called the **σ -field generated by f** , denoted by $\sigma(f)$.

Monotone Class Theorem (I)

Two more primitive classes of sets than the σ -field are the λ -system and the π -system. A λ -**system** is a collection of subsets of Ω that contains Ω and is closed under proper differences and increasing limits. A π -**system** is a nonempty collection of subsets of Ω closed under finite intersections. Note that the power set 2^Ω is clearly both a λ -system and a π -system.

Lemma 1

A collection of subsets of Ω is both a λ -system and a π -system, if and only if it is a σ -field.

Proof: The if part is trivial; To prove the only if part, directly verify the defining properties of σ -field. □

Monotone Class Theorem (II)

Theorem 2 (Monotone class theorem)

If a λ -system $\mathcal{D}$ contains a π -system $\mathcal{C}$, then $\mathcal{D}$ also contains the σ -field generated by $\mathcal{C}$.

Proof: We can clearly assume $\mathcal{D} = \lambda(\mathcal{C})$, the smallest λ -system containing $\mathcal{C}$. It suffices to show $\mathcal{D}$ is a π -system, since then it will be both a λ -system and a π -system, and by Lemma 1 it will be a σ -field containing $\mathcal{C}$ and thus $\sigma(\mathcal{C})$ as well. For any $A, B \in \mathcal{D}$, $A \cap B \in \mathcal{D}$ is clearly true when both A and B are in $\mathcal{C}$ since it is a π -system. We next proceed by two steps. First fix any $B \in \mathcal{C}$, and note $\mathcal{A}_B = \{A \subset \Omega; A \cap B \in \mathcal{D}\}$ contains $\mathcal{C}$ and is a λ -system and so contains $\mathcal{D} = \lambda(\mathcal{C})$. This shows $A \cap B \in \mathcal{D}$ is true for all $B \in \mathcal{C}$ and $A \in \mathcal{D}$. Then fix any $A \in \mathcal{D}$ and note $\mathcal{B}_A = \{B \subset \Omega; B \cap A \in \mathcal{D}\}$ is a λ -system and clearly contains $\mathcal{C}$, so must contain $\mathcal{D} = \lambda(\mathcal{C})$ as well. This shows $A \cap B \in \mathcal{D}$ is true for all $A, B \in \mathcal{D}$. Therefore $\mathcal{D}$ is a π -system. □

Measure and Measure space

A measure is a set function defined on a σ -field that satisfies two fundamental properties we want a probability measure to have, namely non-negativity and σ -additivity.

Formally, a mapping $\mu : \mathcal{F} \rightarrow \overline{\mathbb{R}} \equiv \mathbb{R} \cup \{\pm\infty\}$ is called a **measure** if:

- i) $\mu(\emptyset) = 0$;
- ii) $\mu(A) \geq 0$ for all $A \in \mathcal{F}$; and
- iii) $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$ for any (non-overlapping) sequence $A_i, i = 1, 2, \dots \in \mathcal{F}$ satisfying $i \neq j \Rightarrow A_i \cap A_j = \emptyset$.

The triplet $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**.

Examples of Measures

Well known examples of measure include the Lebesgue measure defined on the Borel σ -field over the real line and counting measures defined on any measurable space.

The **Lebesgue measure** λ is the unique (why?) measure on $\mathcal{B}(\mathbb{R})$ such that $\lambda((a, b)) = b - a$ for all $a \leq b \in \mathbb{R}$.

For any countable set $X = \{x_1, x_2, \dots\} \subset \Omega$, the associated **counting measure** κ_X is defined by $\kappa_X(A) = \sum_{i=1}^{\infty} 1_A(x_i)$ for $A \in \mathcal{F}$, with $1_A(x)$ being the **indicator function** of A which equals 1 if $x \in A$ is true and equals 0 otherwise. For instance, if $X = \mathbb{N} \triangleq \{1, 2, \dots\}$, then $\kappa_X((1.2, 3.1]) = 2$, $\kappa_X(\mathbb{R}) = \infty$, $\kappa_X((0, 1)) = 0$, etc.

Probability Measure and Probability Space (I)

When a measure μ satisfies $\mu(\Omega) < \infty$, it is called a finite measure.

A **probability measure** or simply **probability** P on a measurable space $(\Omega, \mathcal{F})$ is a finite measure that is normalized so that $P(\Omega) = 1$. The triplet $(\Omega, \mathcal{F}, P)$ is called a **probability space**.

The following properties of a probability measure P on $(\Omega, \mathcal{F})$ is easy to verify:

- i) (Monotonicity). $P(A) \leq P(B)$ for all $A, B \in \mathcal{F}$ with $A \subset B$.
- ii) (Continuity). $P(\lim_{n \rightarrow \infty} A_n) = \lim_{n \rightarrow \infty} P(A_n)$, for a monotone sequence $A_1, A_2, \dots \in \mathcal{F}$. Note
 $\lim_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} A_n$ when the A_n are increasing, and
 $\lim_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} A_n$ if they are decreasing.

Also note that for $a_1, a_2, \dots \geq 0$ with $\sum_{i=1}^{\infty} a_i = 1$, and probabilities $P_i, i = 1, 2, \dots$, the set function $\sum_{i=1}^{\infty} a_i P_i$ is also a probability.

Probability Measure and Probability Space (II)

Theorem 3

If two probability measures P and Q agree on a π -system $\mathcal{C}$ of measurable sets, then they also agree on $\sigma(\mathcal{C})$.

By this theorem, when specifying a probability measure on a σ -field, we only need to specify it on a generating π -system.

Proof: Let $\mathcal{D} = \{A \text{ measurable} : P(A) = Q(A)\}$. Then $\mathcal{C} \subset \mathcal{D}$ and $\mathcal{D}$ is a λ -system. Therefore $\sigma(\mathcal{C}) \subset \mathcal{D}$ by the monotone class theorem. □

Some Examples of Probability

Well-known examples of probability measures on $(\mathbb{R}, \mathcal{B})$ include the Gaussian measure $\mathcal{N}_{\mu, \sigma^2}$ with parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ and the Poisson measure $\mathcal{P}_\mu$ with parameter $\mu > 0$.

The former is defined as the unique measure satisfying the requirement

$$\mathcal{N}_{\mu, \sigma^2}((a, b)) = \int_a^b 1/\sqrt{2\pi\sigma^2} \exp\{-(x - \mu)^2/(2\sigma^2)\} dx, \text{ for all } a \leq b.$$

The latter is defined by $\mathcal{P}_\mu(A) = \sum_{i \in A \cap \mathbb{Z}_+} \mu^i \exp(-\mu)/i!$, for all $A \in \mathcal{B}$, where $\mathbb{Z}_+ = \{0\} \cup \mathbb{N} = \{0, 1, 2, \dots\}$ is the set of nonnegative integers.

Measurable Mapping and Measurable Functions (I)

A mapping between two measurable spaces $f : (S, \mathcal{S}) \rightarrow (T, \mathcal{T})$ is called $\mathcal{S}/\mathcal{T}$ -**measurable** or simply **measurable** if the inverse images of all measurable sets are measurable, that is, $f^{-1}B \in \mathcal{S}$ for all $B \in \mathcal{T}$.

It is easy to show the indicator function $1_A(\cdot) = 1\{\cdot \in A\}$ is a measurable function iff A is measurable. Linear combinations of indicator functions of measurable sets are **simple functions**, and can be easily shown to be measurable.

With some effort, it can be shown that the class of measurable functions is closed under the basic finite or countable operations we often encounter, such as linear combination ($\sum_i a_i f_i$), multiplication (fg), division (f/g), supremum ($\sup_n f_n$), infimum ($\inf_n f_n$), limit superior ($\limsup_n f_n$), limit inferior ($\liminf_n f_n$), limit ($\lim_n f_n$), composition ($f \circ g$).

Measurable Mapping and Measurable Functions (II)

Theorem 4 (approximation)

Any nonnegative measurable function is the limit of an increasing sequence of nonnegative simple (measurable) functions.

Proof: For any $f \geq 0$ measurable, we note $f_n = 2^{-n}[2^n f] \wedge n$ is such a sequence, where $[x]$ means taking the integer part of x , and $a \wedge b \triangleq \min\{a, b\}$. □

Theorem 5 (functional representation)

A function $f : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B})$ is measurable with respect to the σ -field generated by another function $g : (\Omega, \mathcal{F}) \rightarrow (T, \mathcal{T})$ if and only if f is a measurable function of g , that is, one can find a measurable $h : (T, \mathcal{T}) \rightarrow (\mathbb{R}, \mathcal{B})$ such that $f(\cdot) = h(g(\cdot))$, or simply $f = h \circ g$.

Proof: If $f = 1_A$ for some $A \in \sigma(g)$, then there exist a measurable B such that $A = g^{-1}B$. Therefore $f = 1_{g^{-1}B} = 1_B \circ g$, and so the result is true with $h = 1_B$. By linearity, the result extends to simple functions. For nonnegative measurable f , we can choose $\sigma(g)$ -measurable simple functions f_n with $0 \leq f_n \uparrow f$, and the associated measurable function $h_1, h_2, \dots$ with $f_n = h_n \circ g$. Then $h = \sup_n h_n$ is measurable and

$$h \circ g = (\sup_n h_n) \circ g = \sup_n (h_n \circ g) = \sup_n f_n = f.$$

By linearity the result extends to general signed measurable functions.

Random Variables and their Distributions

Random variables on a probability space $(\Omega, \mathcal{F}, P)$ are defined as measurable functions, i.e., measurable mappings between $(\Omega, \mathcal{F})$ and $(\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$.

Similarly, **random vectors** are defined as measurable mappings between $(\Omega, \mathcal{F})$ and $(\overline{\mathbb{R}}^n, \mathcal{B}(\overline{\mathbb{R}}^n))$.

If ξ is a random variable defined on $(\Omega, \mathcal{F}, P)$, then the set function $P \circ \xi^{-1}$ that maps Borel measurable sets B to $P(\xi^{-1}B) \in [0, 1]$ is a probability measure on $(\mathbb{R}, \mathcal{B})$. This measure is called the **probability distribution** of ξ .

The function F define by $F(x) = (P \circ \xi^{-1})((-\infty, x])$ is called the **distribution function** of ξ .

Integral and Expectation I

Intuitively, the integral of a measurable function f w.r.t measure ν , $\nu f \equiv \int f d\nu \equiv \int_{\Omega} f(\omega) d\nu(\omega) \equiv \int_{\Omega} f(\omega) \nu(d\omega)$, is the measure weighted sum of the values of f .

More specifically:

- ▶ when $f = 1_A$ for a measurable set A , we define $\nu f = \nu(A)$;
- ▶ when f is a simple function, or $f = \sum_{i=1}^n a_i 1_{A_i}$ for some measurable sets A_i s and nonnegative a_i 's, we define $\nu f = \sum_{i=1}^n a_i \nu(A_i)$ (we use the convention $\infty \cdot 0 = 0$);
- ▶ when $f \geq 0$, there exists a sequence of nonnegative simple functions $f_n \uparrow f$, and we define $\nu f = \lim_n \nu f_n$ (it can be proved this limit does not depend on the choice of the approximation sequence and so this is well-defined);
- ▶ when f is a general measurable function, it can be written as $f = f^+ - f^-$, with $f^+ = f \vee 0 \triangleq \max\{f, 0\}$ and $f^- = (-f) \vee 0$ both nonnegative, and we define $\nu f = \nu f^+ - \nu f^-$.

Integral and Expectation II

When both νf^+ and νf^- are finite, we say f is **integrable** (w.r.t. ν); when at most one of them is ∞ , we say νf exists; when both equal ∞ , we say νf does not exist.

For a random variable ξ defined on the probability space $(\Omega, \mathcal{F}, P)$, its integral with respect to the probability measure P is called its **expectation**, denoted $\mathbb{E}\xi = P\xi$.

Several Important Results About Integrals I

The following are some famous results about integrals we should know. The first is the monotone convergence theorem, which says we can exchange the order of integration and limit if the sequence of functions is nonnegative and increasing.

Theorem 6 (Monotone convergence)

Let $f, f_1, f_2, \dots$ be measurable functions on $(\Omega, \mathcal{F}, \nu)$ with $0 \leq f_n \uparrow f$. Then $\nu f_n \uparrow \nu f$.

Proof: For each n we choose simple measurable function g_{nk} , with $0 \leq g_{nk} \uparrow f_n$ as $k \rightarrow \infty$. Define $h_{nk} = g_{1k} \vee \dots \vee g_{nk}$. Then h_{nk} are all simple measurable functions, and $0 \leq h_{nk} \uparrow f_n$ as $k \rightarrow \infty$. Moreover, $h_{1k} \leq h_{2k} \leq \dots$ for any fixed k . Therefore, $f \geq \lim_{k \rightarrow \infty} h_{kk} \geq \lim_{k \rightarrow \infty} h_{nk} = f_n \uparrow f$, and so $0 \leq h_{kk} \uparrow f$. By the definition and monotonicity of the integral,

$$\nu f = \lim_{k \rightarrow \infty} \nu h_{kk} \leq \lim_{k \rightarrow \infty} \nu f_k \leq \nu f. \quad \square$$

Several Important Results About Integrals II

The second is Fatou's Lemma, which roughly says that integration has a mollifying effects.

Theorem 7 (Fatou's Lemma)

For any measurable functions $f_1, f_2, \dots \geq 0$ on $(\Omega, \mathcal{F}, \nu)$, we have

$$\liminf_{n \rightarrow \infty} \nu f_n \geq \nu \liminf_{n \rightarrow \infty} f_n.$$

Proof: By monotone convergence and monotonicity of integral

$$\nu \liminf_{n \rightarrow \infty} f_n = \nu \lim_{n \rightarrow \infty} \inf_{k \geq n} f_k = \lim_{n \rightarrow \infty} \nu \inf_{k \geq n} f_k \leq \lim_{n \rightarrow \infty} \inf_{m \geq n} \nu f_m = \liminf_{n \rightarrow \infty} \nu f_n. \quad \square$$

The third is the celebrated Lebesgue's dominated convergence theorem, which says we can exchange the order of integration and limit if the convergent sequence of integrands is dominated by an integrable function.

Several Important Results About Integrals III

Theorem 8 (Dominated convergence)

Let $f, f_1, f_2, \dots$ and $g, g_1, g_2, \dots$ be measurable functions on $(\Omega, \mathcal{F}, \nu)$ with $|f_n| \leq g_n$ for all n and such that $f_n \rightarrow f$, $g_n \rightarrow g$ and $\nu g_n \rightarrow \nu g < \infty$. Then $\nu f_n \rightarrow \nu f$.

Note when $g_n \equiv g$, the result reduces to the renowned Lebesgue's dominated convergence theorem.

Proof: Applying Fatou's lemma to the functions $g_n \pm f_n \geq 0$, we have

$$\begin{aligned}\nu g + \liminf_n (\pm \nu f_n) &= \liminf_n \nu(g_n \pm f_n) \\ &\geq \nu \liminf_n (g_n \pm f_n) = \nu(g \pm f) = \nu g \pm \nu f.\end{aligned}$$

Subtracting $\nu g < \infty$ from both sides gives

$$\nu f \leq \liminf_n \nu f_n \leq \limsup_n \nu f_n \leq \nu f. \quad \square$$

Several Important Results About Integrals IV

Another useful result is the substitution formula.

Theorem 9

Consider a measure space $(\Omega, \mathcal{F}, \nu)$, and measurable space $(S, \mathcal{S})$, and two measurable mappings $f : \Omega \rightarrow S$ and $g : S \rightarrow \mathbb{R}$. Then

$$\nu(g \circ f) = (\nu \circ f^{-1})g,$$

whenever either side exists.

Note by this formula, we note

$\mathbb{E}\xi = P\xi = (P \circ \xi^{-1})\iota = \int_{\mathbb{R}} x(P \circ \xi^{-1})(dx)$, where ι is the identity function: $\iota(x) = x$, $\forall x \in \mathbb{R}$.

Proof: Start with $g = 1_A$ for some $A \in \mathcal{S}$, and then extend stepwise to general cases like before. □

Change of Measure and Density

For a measurable function $f \geq 0$ on $(\Omega, \mathcal{F}, \nu)$, define a set function $f \cdot \nu$ on $\mathcal{F}$ by $(f \cdot \nu)A = \nu(1_A f) = \int_A f d\nu, \forall A \in \mathcal{F}$. Then it is easy to check $\mu = f \cdot \nu$ is still a measure on $(\Omega, \mathcal{F})$. Here f is referred to as the ν -density of μ , and in notation, $f = \frac{d\mu}{d\nu}$, or $f(\omega) = \frac{\mu(d\omega)}{\nu(d\omega)}$. For a random variable ξ , if there is a measurable function $f \geq 0$ such that $P \circ \xi^{-1} = f \cdot \lambda$ with λ being the Lebesgue measure, then ξ is said to be an absolutely continuous random variable, and f is also called its **density function**.

If f is such that $P \circ \xi^{-1} = f \cdot \kappa$, with κ being the counting measure associated with a countable set of $\mathbb{R}$, then ξ is said to be a discrete random variable, and f , the κ -density of $P \circ \xi^{-1}$, is also known as its **(probability) mass function**.

Theorem 10 (chain rule)

For any measure space $(\Omega, \mathcal{F}, \nu)$ and measurable functions $f : \Omega \rightarrow \mathbb{R}_+$ and $g : \Omega \rightarrow \mathbb{R}$, $\nu(fg) = (f \cdot \nu)g$, whenever either side exists.

Proof: The same old trick; Start with indicator functions. ▶

Product Measure and Fubini's Theorem I

We now state the Fubini's theorem but won't prove it (cf. Theorem 1.27 of Kallenberg (2002) for a proof). Note for two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$, we can define a σ -field on the product space $S \times T$, $\mathcal{S} \otimes \mathcal{T} = \sigma(\{A \times T; A \in \mathcal{S}\} \cup \{S \times B; B \in \mathcal{T}\})$, called the **product σ -algebra**.

We call a measure σ -finite if the whole space can be partitioned into countably many disjoint measurable sets all with finite measure (under this measure).

Product Measure and Fubini's Theorem II

Theorem 11 (Fubini's Theorem)

For any two σ -finite measure spaces $(S, \mathcal{S}, \mu)$ and $(T, \mathcal{T}, \nu)$, there exists a unique measure $\mu \otimes \nu$ on $(S \times T, \mathcal{S} \otimes \mathcal{T})$ satisfying

$$(\mu \otimes \nu)(A \times B) = \mu A \cdot \nu B, \quad \forall A \in \mathcal{S}, B \in \mathcal{T}.$$

Furthermore, for any measurable function $f : S \times T \rightarrow \mathbb{R}_+$,

$$(\mu \otimes \nu)f = \int \mu(ds) \int f(s, t) \nu(dt) = \int \nu(dt) \int f(s, t) \mu(ds).$$

The last relation remains valid for any $\mu \otimes \nu$ -integrable function $f : S \times T \rightarrow \mathbb{R}$.

Two Integral Inequalities I

Given a measure space $(\Omega, \mathcal{F}, \nu)$ and a $p > 0$, we define the L^p -norm of f by $\|f\|_p = (\nu|f|^p)^{1/p}$.

Theorem 12 (Hölder and Minkowski inequalities)

For any two measurable functions f and g on some measure space $(\Omega, \mathcal{F}, \nu)$, we have

- (i) $\nu|fg| \leq \|f\|_p \|g\|_q$ for all $p, q > 0$ with $p^{-1} + q^{-1} = 1$,
- (ii) $\|f + g\|_p^{p \wedge 1} \leq \|f\|_p^{p \wedge 1} + \|g\|_p^{p \wedge 1}$.

Proof: (i) Clearly, we only need to consider the case with $\|f\|_p = \|g\|_q = 1$. Since $p^{-1} + q^{-1} = 1$ means $(p-1)(q-1) = 1$, and so $y = x^{p-1}$ and $x = y^{q-1}$ are equivalent for $x, y \geq 0$. By the elementary relation (draw a diagram)

$$|fg| \leq \int_0^{|f|} x^{p-1} dx + \int_0^{|g|} y^{q-1} dy = p^{-1}|f|^p + q^{-1}|g|^q,$$

Two Integral Inequalities II

and so

$$\nu|fg| \leq p^{-1}\nu|f|^p + q^{-1}\nu|g|^q = p^{-1} + q^{-1} = 1.$$

(ii) When $p \leq 1$ the relation follows from the concavity of x^p on $\mathbb{R}_+$. For $p > 1$, by $|x + y| \leq |x| + |y|$ and the Hölder inequality,

$$\begin{aligned}\|f + g\|_p^p &\leq \nu(|f||f + g|^{p-1}) + \nu(|g||f + g|^{p-1}) \\ &\leq \|f\|_p \| |f + g|^{p-1} \|_{p/(p-1)} + \|g\|_p \| |f + g|^{p-1} \|_{p/(p-1)}.\end{aligned}$$

Now divide both sides by $\| |f + g|^{p-1} \|_{p/(p-1)} = \|f + g\|_p^{p-1}$. □

Note setting $p = q = 2$ in the Hölder inequality gives the

Cauchy-Buniakovsky or **Schwarz** inequality,

$$\|fg\|_1 \leq \|f\|_2 \|g\|_2.$$

By Hölder inequality, the existence of an absolute moment implies existence of all lower order absolute moments, since

$$\mathbb{E}[|\xi|^p] \leq \| |\xi|^p \|_{q/p} \|1\|_{1/(1-p/q)} = \mathbb{E}[|\xi|^q]^{p/q}, \quad \forall 0 < p < q.$$

Jensen's Inequality I

A random vector $\xi = (\xi_1, \dots, \xi_d)$ is said to be integrable if integrability holds for every component ξ_k , in which case we write $\mathbb{E}\xi \triangleq (\mathbb{E}\xi_1, \dots, \mathbb{E}\xi_d)$. Another often used inequality is the Jensen's inequality. Recall that a function f defined on a linear space X is said to be convex if

$$f(px + (1-p)y) \leq pf(x) + (1-p)f(y) \text{ for all } x, y \in X \text{ and } p \in [0, 1].$$

Theorem 13 (Jensen's inequality)

For any integrable random vector ξ in $\mathbb{R}^d$ and convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, we have

$$\mathbb{E}f(\xi) \geq f(\mathbb{E}\xi).$$

Note this result also implies

$$\mathbb{E}|\xi|^q = \mathbb{E}(|\xi|^p)^{q/p} \geq (\mathbb{E}[|\xi|^p])^{q/p} \implies \|\xi\|_p \leq \|\xi\|_q \text{ for } p \leq q.$$

Jensen's Inequality II

Proof: By the property of convex functions, for any $s \in \mathbb{R}^d$ there exists a supporting affine function $h_s(x) = ax + b$ with $f \geq h_s(x)$ and $f(s) = h_s(s)$. Taking $s = \mathbb{E}\xi$ gives

$$\mathbb{E}f(\xi) \geq \mathbb{E}h_s(\xi) = h_s(\mathbb{E}\xi) = f(\mathbb{E}\xi). \quad \square$$

If a random variable has finite second moment, its **variance** is defined as $\text{var}(\xi) = \mathbb{E}(\xi - \mathbb{E}\xi)^2 = \mathbb{E}\xi^2 - (\mathbb{E}\xi)^2$.

Independence

Two events $A, B \in \mathcal{F}$ are said to be **independent** if $P(AB) \triangleq P(A \cap B) = P(A)P(B)$. Two collections of measurable sets $\mathcal{C}_1$ and $\mathcal{C}_2$ are called independent, if any $A \in \mathcal{C}_1$ and $B \in \mathcal{C}_2$ are independent. Two random variables are said to be independent if the two σ -fields generated by them are independent. A measurable set and a collection of measurable sets are called independent if the measurable set is independent of any event in the collection. A random variable ξ on $(\Omega, \mathcal{F}, P)$ and a collection $\mathcal{A}$ of measurable sets are independent if $\sigma(\xi)$ and $\mathcal{A}$ are independent.

Some simple implications of these definitions include: trivial events (i.e. measurable sets with probability zero or one) are independent of all other measurable sets, or random variables, or collections of measurable sets; the trivial sigma field $\{\emptyset, \Omega\}$ is independent of all measurable sets, random variables, sigma-fields, etc; if two random variables ξ and η are independent and $\mathbb{E}[|\xi\eta|] < \infty$ then $\mathbb{E}[\xi\eta] = \mathbb{E}[\xi]\mathbb{E}[\eta]$ (This last one can be proved using the “old trick”).

Probability Kernels I

Given two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$, a mapping $\mu : S \times \mathcal{T} \rightarrow \overline{\mathbb{R}}_+$ is called a **(probability) kernel** from S to T if the function $\mu_s(B) = \mu(s, B)$ is $\mathcal{S}$ -measurable in $s \in S$ for fixed $B \in \mathcal{T}$ and a (probability) measure in $B \in \mathcal{T}$ for fixed $s \in S$.

Any kernel μ determines an associated operator that maps suitable functions $f : T \rightarrow \mathbb{R}$ into their integrals $\mu f(s) = \int \mu(s, dt)f(t)$.

Kernels play an important role in probability theory, where they may appear in the guises of random measures, conditional distributions, and **Markov transition functions**.

Let $\mathcal{P}(T)$ denote the set of probability measures on T , endowed with the σ -field induced (generated) by the mappings $\pi_B : \mu \mapsto \mu B, B \in \mathcal{T}$.

Probability Kernels II

Theorem 14

Fix two measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$, a π -system $\mathcal{C}$, with $\sigma(\mathcal{C}) = \mathcal{T}$, and a family $\mu = \{\mu_s; s \in S\}$ of probability measures on T . Then these conditions are equivalent:

- (i) μ is a probability kernel from S to T ;
- (ii) μ is a measurable mapping from S to $\mathcal{P}(T)$;
- (iii) $s \mapsto \mu_s(B)$ is a measurable mapping from S to $[0, 1]$ for every $B \in \mathcal{C}$.

Proof: Since $\pi_B : \mu \mapsto \mu B$ is measurable on $\mathcal{P}(T)$ for every $B \in \mathcal{T}$, condition (ii) implies (iii) by the measurability of composition of measurable mappings. Furthermore, (iii) implies (i) by a straightforward application of the monotone class theorem. Finally, under (i) we have $\mu^{-1}\pi_B^{-1}[0, x] \in \mathcal{S}$ for all $B \in \mathcal{T}$ and $x \geq 0$, and (ii) follows from the fact that $\{\pi_B^{-1}[0, x]; B \in \mathcal{T}, x \geq 0\}$ generate the σ -field on $\mathcal{P}(T)$. □

Probability Kernels III

Fix three measurable spaces $(S, \mathcal{S})$, $(T, \mathcal{T})$, and $(U, \mathcal{U})$. Let μ and ν be two probability kernels from S to T , and from $S \times T$ to U , respectively. We define their product:

$$(\mu \otimes \nu)(s, B) = \int \mu(s, dt) \int \nu(s, t, du) 1_B(t, u), \quad B \in \mathcal{T} \otimes \mathcal{U}.$$

Then $\mu \otimes \nu$ is a kernel from S to $T \times U$. For any measurable function $f \geq 0$ on $T \times U$, we have

$$(\mu \otimes \nu)_s f = \int \mu(s, dt) \int \nu(s, t, du) f(t, u), \quad s \in S,$$

or $(\mu \otimes \nu)f = \mu(\nu f)$. By iteration, we may combine any kernels μ_k from $S_0 \times \cdots \times S_{k-1}$ to S_k , $k = 1, \dots, n$, into a kernel $\mu_1 \otimes \cdots \otimes \mu_n$ from S_0 to $S_1 \times \cdots \times S_n$, given by

$$(\mu_1 \otimes \cdots \otimes \mu_n)f = \mu_1(\mu_2(\cdots(\mu_n f) \cdots))$$

for any measurable function $f \geq 0$ on $S_1 \times \cdots \times S_n$.

Probability Kernels IV

For kernels μ_k from S_{k-1} to S_k , $k = 1, \dots, n$, the composition $\mu_1 \cdots \mu_n$ is defined as a kernel from S_0 to S_n given by

$$\begin{aligned} (\mu_1 \cdots \mu_n)_s B = & \int \mu_1(s, ds_1) \int \mu_2(s_1, ds_2) \cdots \\ & \cdots \int \mu_{n-1}(s_{n-2}, ds_{n-1}) \mu_n(s_{n-1}, B), \end{aligned}$$

for any measurable $B \subset S_n$.

For a probability measure ν on S_0 , and probability kernels μ_k from S_{k-1} to S_k , $k = 1, \dots, n$, the following is a probability measure on S_n :

$$(\nu \mu_1 \cdots \mu_n) B = \int \nu(ds) (\mu_1 \cdots \mu_n)_s(B).$$

Conditional expectation I

If B is measurable set in $(\Omega, \mathcal{F}, P)$ with $P(B) > 0$, then the set function $P(\cdot \cap B)/P(B)$ can be verified to be a probability measure on $\mathcal{F}$, and is called the conditional probability measure given B , denoted by $P(\cdot|B)$. The conditional expectation of a random variable ξ given B is defined as its expectation with respect to the conditional probability measure given B : $\mathbb{E}[\xi|B] = \int \xi(\omega)P(d\omega|B)$. It can be verified (using the “old trick”) that $\mathbb{E}[\xi|B] = \mathbb{E}[\xi 1_B]/P(B)$.

The conditional expectation so defined varies with the conditioning measurable set (or **event**) B . For instance, if $P(B) \in (0, 1)$, then $\mathbb{E}[\xi|B]$ and $\mathbb{E}[\xi|B^c]$ are both well defined and could be two different values. It is handy to define a function from which we can tell the expected value of ξ according to the “status of nature” ω ,

$$\eta(\omega) = \mathbb{E}[\xi|B]1_B(\omega) + \mathbb{E}[\xi|B^c]1_{B^c}(\omega).$$

Conditional expectation II

This is a random variable which equals $\mathbb{E}[\xi|B]$ when $\omega \in B$ (B happens) and equals $\mathbb{E}[\xi|B^c]$ when $\omega \in B^c$ or $\omega \notin B$ (B does not happen). In other words, this random variable tells us the expected value of ξ given our knowledge of the event B (whether it happens or not). In fact, it is easy to verify that η is measurable with respect to the σ -field generated by the event B , $\sigma(B) \triangleq \sigma(\{B\}) = \{\emptyset, \Omega, B, B^c\}$, and satisfies the requirement that $\mathbb{E}[\eta; A] \triangleq \mathbb{E}[\eta 1_A] = \mathbb{E}[\xi 1_A]$ for all $A \in \sigma(B)$. Therefore, this random variable η is also called the conditional expectation of ξ given the σ -field $\sigma(B)$.

Generally, the **conditional expectation** of an integrable random variable ξ on $(\Omega, \mathcal{F}, P)$ given a σ -field $\mathcal{G} \subset \mathcal{F}$, denoted by $\mathbb{E}[\xi|\mathcal{G}] \equiv \mathbb{E}^{\mathcal{G}}\xi$, is defined as any function η such that

- (i) η is $\mathcal{G}$ -measurable;
- (ii) $\mathbb{E}[\eta; A] = \mathbb{E}[\xi; A], \forall A \in \mathcal{G}$.

Conditional expectation III

NB: The existence of conditional expectation is by the Radon-Nikodym theorem, which says that if two measures μ and ν on the same measurable space is such that $\mu A = 0 \Rightarrow \nu A = 0$ for all measurable sets A (ν is said to be **dominated** by μ ; or in notation, $\nu \ll \mu$), then there exists a measurable function f such that $\nu = f \cdot \mu$; see Theorem 2.10 of Kallenberg if you are interested in more details. The conditional expectation is not unique, but any two versions can only differ on a set of P -measure 0, and so is almost surely unique.

The conditional expectation of a random variable ξ given another random variable ζ is defined by $\mathbb{E}[\xi|\zeta] \triangleq \mathbb{E}[\xi|\sigma(\zeta)]$. By the functional representation theorem (Theorem 5), there exists a measurable function h such that $\mathbb{E}[\xi|\zeta] = h \circ \zeta$.

The **conditional probability** of an event $B \in \mathcal{F}$ given a sigma-field $\mathcal{G} \subset \mathcal{F}$, is defined by $P(B|\mathcal{G}) = \mathbb{E}[1_B|\mathcal{G}]$.

Properties of conditional expectation I

The following are some properties of conditional expectations.

Recall that a relation holds a.s. (almost surely) means by definition that the relation holds outside a measurable set of probability 0.

The notation a.e. (almost everywhere) is similarly defined.

Theorem 15

Let ξ and η be integrable r.v.s on $(\Omega, \mathcal{F}, P)$ and $\mathcal{G}, \mathcal{H}$ be sub- σ -fields.

- i. (linearity) $\mathbb{E}^{\mathcal{G}}[\cdot]$ is a linear operator;
- ii. (no information is no information) $\mathbb{E}[\xi | \{\emptyset, \Omega\}] = \mathbb{E}[\xi]$ a.s.;
- iii. (irrelevant conditions can be dropped) $\mathbb{E}^{\mathcal{G}}\xi = \mathbb{E}\xi$ a.s. if ξ and $\mathcal{G}$ are independent;
- iv. (positivity) $\xi \geq 0 \Rightarrow \mathbb{E}^{\mathcal{G}}\xi \geq 0$ a.s.;
- v. (contractivity) $\mathbb{E}|\mathbb{E}^{\mathcal{G}}\xi| \leq \mathbb{E}|\xi|$;
- vi. (monotone convergence)* $0 \leq \xi_n \uparrow \xi \Rightarrow \mathbb{E}^{\mathcal{G}}\xi_n \uparrow \mathbb{E}^{\mathcal{G}}\xi$ a.s. ;
- vii. (pull-out) Assume $\xi\eta$ is integrable. Then ξ is $\mathcal{G}$ measurable $\Rightarrow \mathbb{E}^{\mathcal{G}}\xi\eta = \xi\mathbb{E}^{\mathcal{G}}\eta$;

Properties of conditional expectation II

Theorem 15

- viii. (tower/telescoping property) $\mathbb{E}^{\mathcal{G}}\mathbb{E}^{\mathcal{H}}\xi = \mathbb{E}^{\mathcal{G}}\xi$ a.s. for $\mathcal{G} \subset \mathcal{H}$.
- ix. (Jensen's inequality) If f is a convex function on $\mathbb{R}$ such that $f(\xi)$ is integrable, then

$$f(\mathbb{E}[\xi|\mathcal{G}]) \leq \mathbb{E}[f(\xi)|\mathcal{G}], \text{ a.s.}$$

- x. (Fatou's lemma) If $\xi_n, n = 1, 2, \dots$ is a sequence of nonnegative random variables, Then

$$\mathbb{E}[\liminf_n X_n|\mathcal{G}] \leq \liminf_n \mathbb{E}[X_n|\mathcal{G}].$$

Properties of conditional expectation III

Proof: (i)-(iii) are easy (can be verified by definition).

(iv) Let $A = \{\mathbb{E}^{\mathcal{G}}\xi < 0\}$, and $A_n = \{\mathbb{E}^{\mathcal{G}}\xi \leq -1/n\}$. Then $A = \bigcup_{n=1}^{\infty} A_n$. By $\xi \geq 0$, we have

$0 \leq \mathbb{E}[\xi 1_{A_n}] = \mathbb{E}[(\mathbb{E}^{\mathcal{G}}\xi) 1_{A_n}] \leq -1/n P(A_n)$, which means $P(A_n) = 0$. So $P(A) = \lim_n P(A_n) = 0$.

(v) Let $A = \{\mathbb{E}^{\mathcal{G}}\xi \geq 0\}$. Then

$$\begin{aligned}\mathbb{E}|\mathbb{E}^{\mathcal{G}}\xi| &= \mathbb{E}[|\mathbb{E}^{\mathcal{G}}\xi|; A] + \mathbb{E}[|\mathbb{E}^{\mathcal{G}}\xi|; A^c] = \mathbb{E}[\mathbb{E}^{\mathcal{G}}\xi; A] + \mathbb{E}[-\mathbb{E}^{\mathcal{G}}\xi; A^c] \\ &= \mathbb{E}[\xi; A] + \mathbb{E}[-\xi; A^c] \leq \mathbb{E}[|\xi|; A] + \mathbb{E}[|\xi|; A^c] = \mathbb{E}[|\xi|].\end{aligned}$$

(vi) If $0 \leq \xi_n \uparrow \xi$ we have $0 \leq \mathbb{E}^{\mathcal{G}}\xi_n \uparrow$ a.s. by linearity and positivity. Let N be the exception set with $P(N) = 0$ outside of which the relation is true everywhere. Let η be the random variable which equals the limit of $\mathbb{E}^{\mathcal{G}}\xi_n$ on N^c and equals 0 on N .

Properties of conditional expectation IV

So $0 \leq \mathbb{E}^{\mathcal{G}}\xi_n \uparrow \eta$, and $0 \leq 1_A \mathbb{E}^{\mathcal{G}}\xi_n \uparrow 1_A \eta$ for all $A \in \mathcal{G}$. By monotone convergence

$$\mathbb{E}[\eta; A] = \lim_n \mathbb{E}[\mathbb{E}^{\mathcal{G}}\xi_n; A] = \lim_n \mathbb{E}[\xi_n; A] = \mathbb{E}[\lim_n \xi_n; A] = \mathbb{E}[\xi; A], \quad \forall A \in \mathcal{G}.$$

So $\eta = \mathbb{E}^{\mathcal{G}}\xi$.

(vii) The relation is clearly true if ξ is an indicator function. By linearity and monotone convergence it extends to simple, nonnegative measurable, and then general integrable functions.

(viii) Note $\mathbb{E}^{\mathcal{G}}\xi$ is $\mathcal{G}$ -measurable, and $\forall A \in \mathcal{G} \subset \mathcal{H}$, $\mathbb{E}[\mathbb{E}^{\mathcal{G}}\xi; A] = \mathbb{E}[\xi; A] = \mathbb{E}[\mathbb{E}^{\mathcal{H}}\xi; A]$. So, $\mathbb{E}[\mathbb{E}^{\mathcal{H}}\xi | \mathcal{G}] = \mathbb{E}[\xi | \mathcal{G}]$.

(ix) By a property of convex functions (see e.g. p. 61 of William's Probability with Martingales), there exist (a_n, b_n) , $n = 1, 2, \dots$, such that

$$f(x) = \sup_n (a_n x + b_n), \quad x \in \mathbb{R}.$$

Properties of conditional expectation V

So $f(\mathbb{E}[\xi|\mathcal{G}]) \geq a_n \mathbb{E}[\xi|\mathcal{G}] + b_n$ a.s. for all n , and thus

$$\mathbb{E}[f(\xi)|\mathcal{G}] \geq \sup_n (a_n \mathbb{E}[\xi|\mathcal{G}] + b_n) = f(\mathbb{E}[\xi|\mathcal{G}]), \text{ a.s.}$$

(x): Use part (vi).

